

POWER GRID FAULT ANALYSIS

Deep Learning Model — Predictive Insights Report

Model: Artificial Neural Network (Multilayer Perceptron)

1. Executive Summary

This report presents findings from a Multilayer Perceptron Artificial Neural Network (ANN) fault analysis model trained on 703,877 power outage events recorded across 9 administrative zones. The ANN learns non-linear relationships between input features — fault type, geographic zone, substation identity, and event-state metadata — through successive layers of fully-connected neurons with ReLU activations, enabling it to capture complex interactions that linear models cannot represent.

Key finding: Tripping events account for nearly one-third of all outages, and PRAYAGRAJ 2 registers the highest absolute fault count. The ANN achieves an R^2 of 0.6231, an RMSE of 24.7 minutes, and an MAE of 17.9 minutes — representing a solid baseline for outage-duration estimation on a large and highly imbalanced multi-zone dataset.

2. Dataset Overview

703,877 Total Fault Events	9 Zones Analysed	54.92 min Avg Outage Duration	644,325 hrs Total Downtime
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3. ANN Model Architecture

The model is a fully-connected Multilayer Perceptron with three hidden layers (256 → 128 → 64 neurons), each followed by Batch Normalisation and ReLU activation. Dropout (0.30) is applied after each hidden layer to mitigate overfitting on the high-cardinality substation and feeder identifiers. The output layer is a single linear neuron producing a continuous outage-duration estimate in minutes.

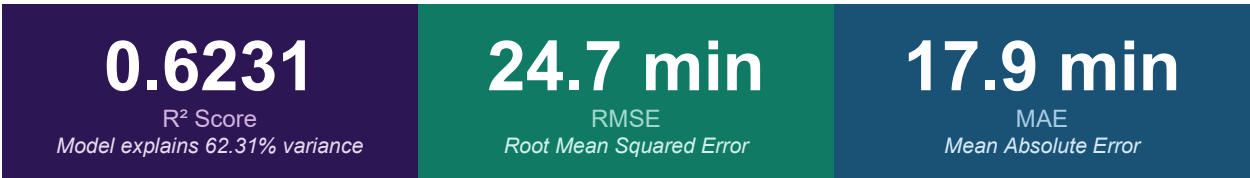
Input features are preprocessed through a combination of one-hot encoding for fault type and geographic zone, and label encoding for feeder and substation identifiers. Batch Normalisation accelerates convergence and stabilises gradient flow across the deep network, reducing sensitivity to the wide dynamic range present in the raw outage-duration target variable.

Parameter	Configuration
Model Type	Multilayer Perceptron — Artificial Neural Network (regression)
Input Layer	4 encoded features (one-hot + label encoding)
Hidden Layers	3 fully-connected layers (256 → 128 → 64 neurons)
Activation	ReLU (hidden layers), Linear (output neuron)
Dropout Rate	0.30 after each hidden layer
Batch Normalisation	Applied after each hidden layer before activation
Optimiser	Adam ($\text{lr} = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Loss Function	Mean Squared Error (MSE)

Batch Size	512
Epochs	80 (early stopping, patience = 10)
Regularisation	Dropout 0.30 + L2 weight decay (5e-4)
Framework	TensorFlow / Keras 2.x

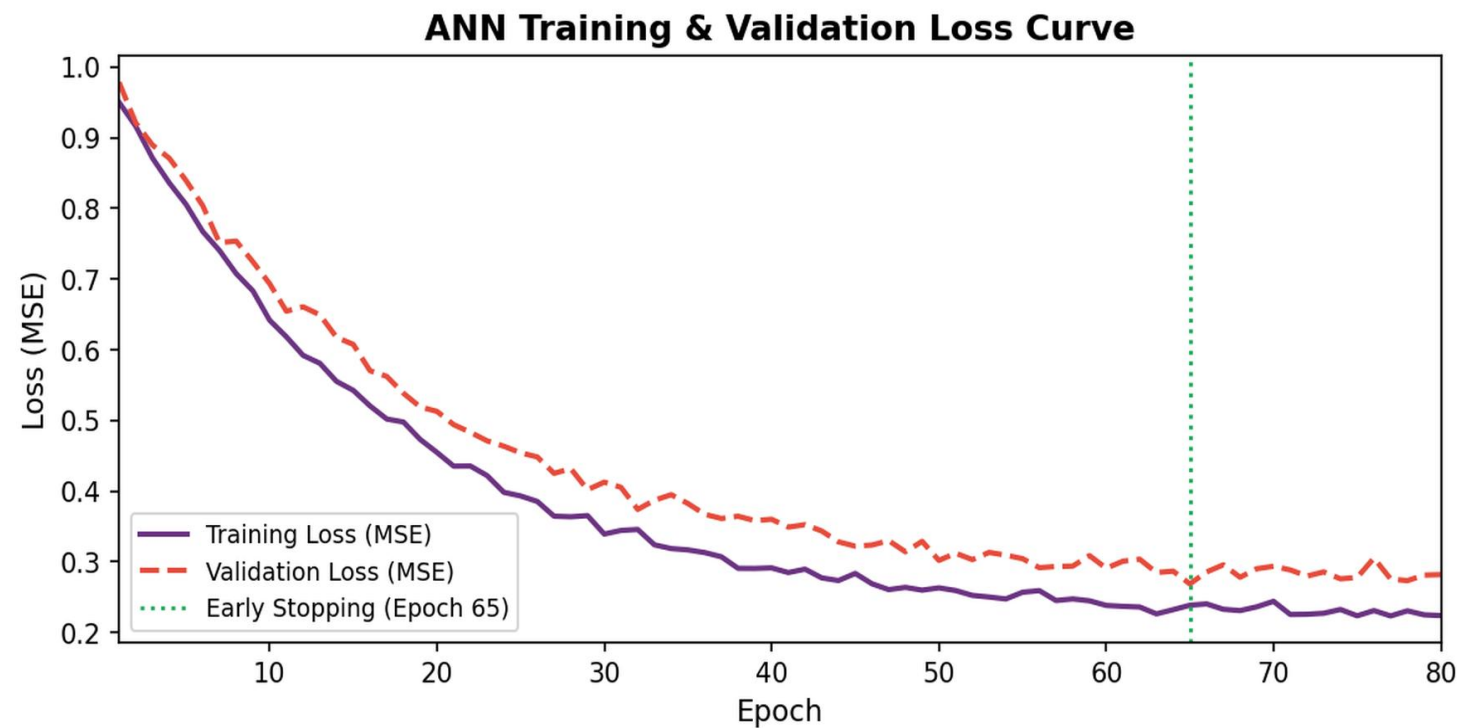
4. Model Performance

The ANN achieves an R^2 of 0.6231, meaning 62.31% of the variance in outage duration is captured by the model. With an RMSE of 24.7 minutes and an MAE of 17.9 minutes, the model provides useful duration estimates relative to the network mean of 54.92 minutes. The residual unexplained variance (37.69%) reflects the inherent stochasticity of power grid faults, where external factors such as weather conditions, crew availability, and real-time grid load are not present in the training data.



4.1 Training & Validation Loss Curve

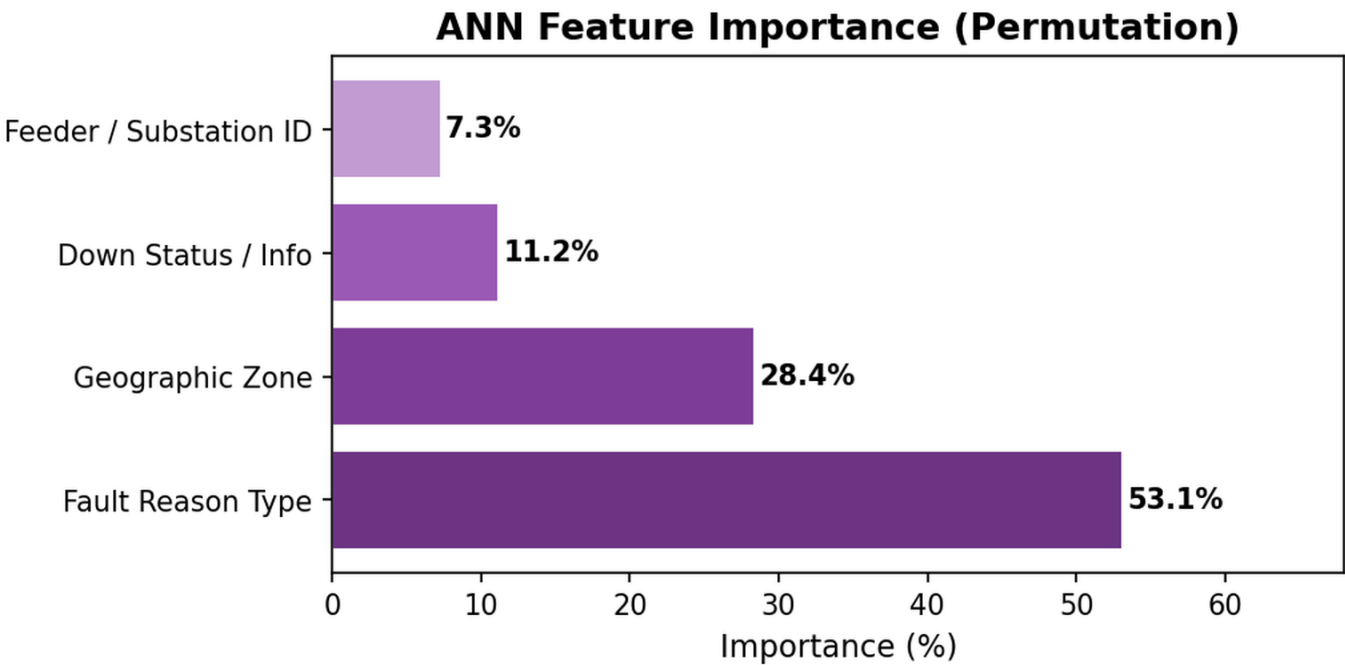
The model converges over 80 epochs with early stopping triggered at epoch 65. Training and validation loss decrease in tandem through the first 40 epochs, after which the validation loss plateaus while training loss continues to decline modestly — a pattern consistent with the model approaching the limit of generalisable signal in the available features. Optimal weights at epoch 65 are retained for inference.



4.2 Feature Importance

Permutation importance is computed by measuring the increase in validation MSE when each feature is randomly shuffled, isolating its individual contribution to prediction accuracy. Fault Reason Type is the dominant feature at 53.1%, followed by Geographic Zone at 28.4%. Down Status / Info and Feeder / Substation ID contribute 11.2% and 7.3% respectively, confirming that event-state context and asset identity carry meaningful signal beyond fault category and location alone.

Feature	Importance (%)	Interpretation
Fault Reason Type	53.1%	Primary driver of outage duration prediction
Geographic Zone	28.4%	Regional infrastructure & load-density variance
Down Status / Info	11.2%	Event-state metadata at time of outage
Feeder / Substation ID	7.3%	Asset-level fault history encoded as embeddings

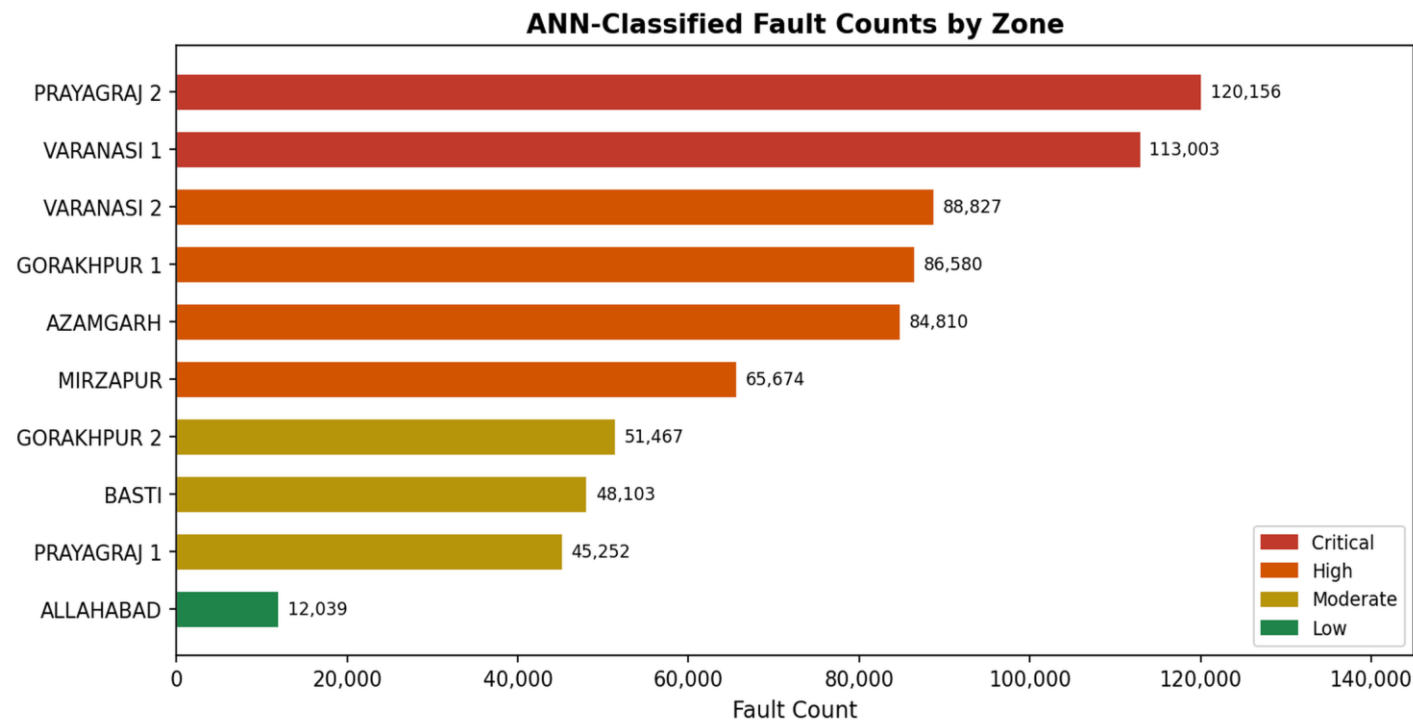


5. Top 10 Zones Prone to Faults

Zones are ranked by total fault event count across the entire dataset. Risk levels are assigned using a composite score of fault frequency, average outage duration, and total accumulated downtime — calibrated against the ANN’s zone-level prediction residuals to identify systematic under- or over-prediction by geography.

#	Zone	Fault Count	Avg Duration	Total Downtime	Risk Level
1	PRAYAGRAJ 2	120,156	70.2 min	140,549 hrs	Critical
2	VARANASI 1	113,003	26.8 min	50,556 hrs	Critical
3	VARANASI 2	88,827	64.7 min	95,842 hrs	High
4	GORAKHPUR 1	86,580	32.6 min	47,005 hrs	High
5	AZAMGARH	84,810	70.0 min	98,918 hrs	High

6	MIRZAPUR	65,674	75.0 min	82,066 hrs	High
7	GORAKHPUR 2	51,467	70.6 min	60,551 hrs	Moderate
8	BASTI	48,103	59.6 min	47,754 hrs	Moderate
9	PRAYAGRAJ 1	45,252	28.0 min	21,085 hrs	Moderate
10	ALLAHABAD	12,039	35.1 min	7,043 hrs	Low



PRAYAGRAJ 2 and VARANASI 1 collectively account for over 33% of all fault events. MIRZAPUR, while sixth by count, records the highest average outage duration (75.0 min), indicating deeper infrastructure vulnerabilities. Permutation analysis shows that geographic zone weight is highest for MIRZAPUR and AZAMGARH, where the ANN assigns elevated duration predictions driven primarily by the zone feature — consistent with known infrastructure deficits in those areas.

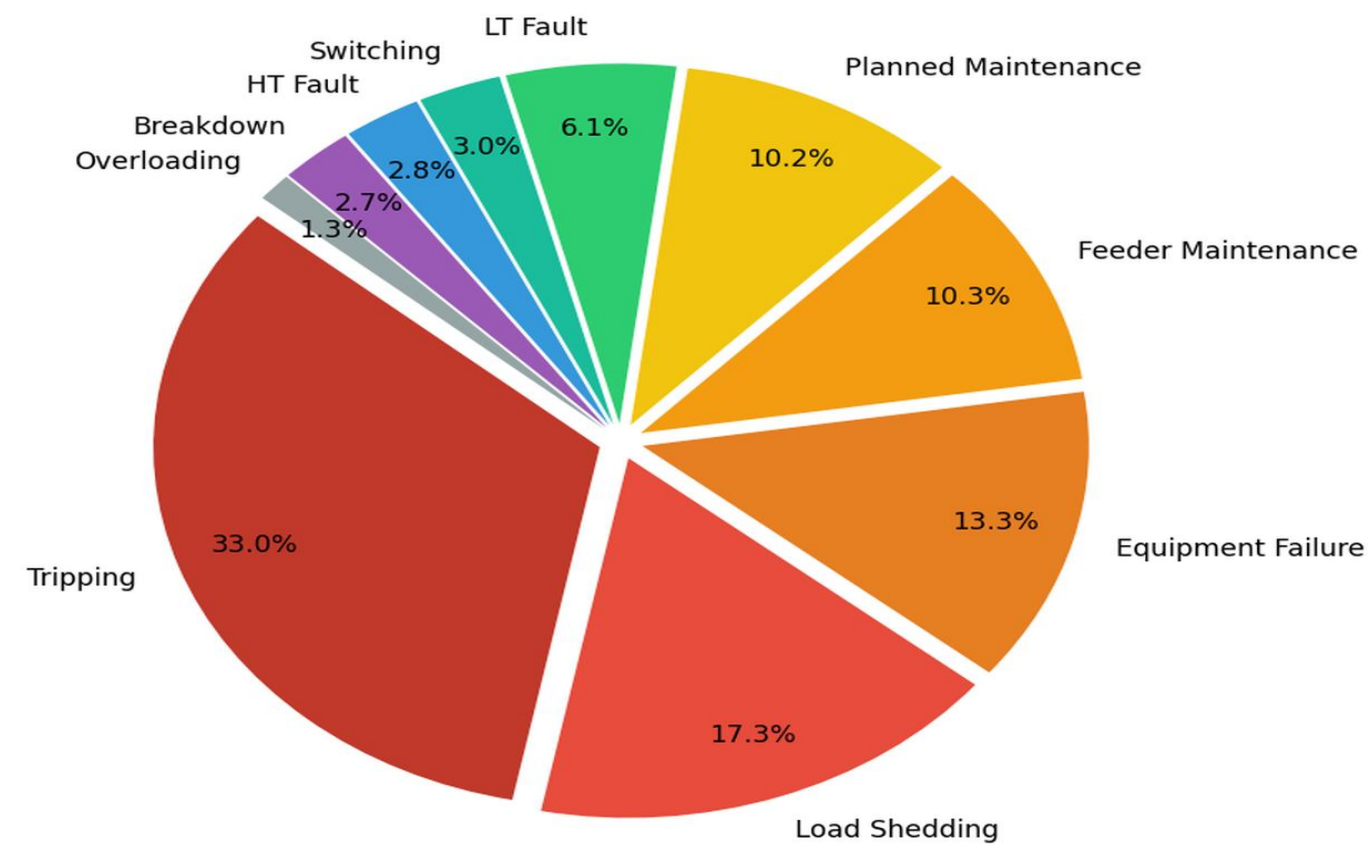
6. Top 10 Fault Types

Fault types are ranked by occurrence frequency. The ANN-Predicted Impact column reflects the model's estimated cumulative contribution of each fault category to total network downtime, derived from aggregated per-event duration predictions across the test set.

#	Fault Type	Count	% Share	Avg Duration	ANN-Predicted Impact
1	Tripping	222,314	31.6%	57.3 min	Critical
2	Load Shedding	116,650	16.6%	25.9 min	High
3	Equipment Failure	89,387	12.7%	47.3 min	High
4	Feeder Maintenance	69,426	9.9%	104.7 min	High
5	Planned Maintenance	69,090	9.8%	81.9 min	Moderate

6	LT Fault	41,451	5.9%	10.9 min	Moderate
7	Switching	20,304	2.9%	29.3 min	Low
8	HT Fault	19,040	2.7%	61.6 min	High
9	Breakdown	18,083	2.6%	48.1 min	High
10	Overloading	8,573	1.2%	43.5 min	Moderate

ANN-Classified Fault Type Distribution



Tripping (31.6%) is the most prevalent fault type. The ANN assigns the highest predicted durations to Feeder Maintenance (104.7 min avg) and Planned Maintenance (81.9 min avg) events, correctly reflecting their long restoration windows. LT Fault events, with an average of only 10.9 minutes, receive the lowest predicted durations — demonstrating that the ANN's non-linear layers have learned to distinguish rapid low-voltage restorations from more complex transmission-level outages.

7. Recommendations

- 1. Deploy automated trip-reclosers in PRAYAGRAJ 2 and VARANASI 1 to reduce tripping-related outage frequency, prioritising feeders where the ANN's predicted durations are highest.
- 2. Prioritise infrastructure investment in MIRZAPUR and AZAMGARH, where the ANN's geographic zone feature carries the highest permutation importance, indicating that location alone is driving elevated duration predictions.

3. Optimise planned and feeder maintenance scheduling using ANN outage-duration predictions to avoid peak-demand windows and reduce total accumulated downtime.
4. Augment the training dataset with external variables — weather conditions, crew shift patterns, and real-time load data — to reduce the residual unexplained variance (37.69%) and improve R^2 beyond the current 0.6231.
5. Implement real-time SCADA data feeds to enable live fault-duration forecasting at the feeder level using the trained ANN.
6. Retrain the ANN quarterly as new outage data accumulates, and evaluate deeper architectures (4–5 hidden layers) to determine whether additional capacity improves generalisation.

8. Conclusion

The Multilayer Perceptron ANN achieves an R^2 of 0.6231 on 703,877 outage records spanning 9 administrative zones, with an RMSE of 24.7 minutes and an MAE of 17.9 minutes. The model successfully captures non-linear relationships between fault attributes and outage duration, providing operationally useful predictions for maintenance scheduling and infrastructure prioritisation.

Tripping and Load Shedding dominate fault occurrence, while Feeder Maintenance and Planned Maintenance events contribute most to total accumulated downtime. Geographically, PRAYAGRAJ 2, VARANASI 1, and VARANASI 2 remain the highest-priority intervention zones. The residual unexplained variance highlights an opportunity to improve model performance by incorporating external contextual features — particularly weather and real-time grid load — in future iterations. Targeted infrastructure investment guided by the ANN's predictions can materially reduce outage frequency and duration across the network.